

2.5.1

题目属于非正弦交流电. 将信号拆分为杂波解决.

对于直流分量: $I=0$. $\therefore U_{00} = U_{I0} = 7.85V$

对于 $10\sin\omega t$: $X_{L1} = \omega L = 1000\Omega$ $X_{C1} = \frac{1}{\omega C} = 1000\Omega$
 $\therefore U_{01} = U_{I1} \frac{j(X_{L1} - X_{C1})}{R + j(X_{L1} - X_{C1})} = 0$

对于 $3.33\sin 3\omega t$: $X_{L2} = 3\omega L = 3000\Omega$ $X_{C2} = \frac{1}{3\omega C} = \frac{1000}{3}\Omega$

$U_{02} = U_{I2} \frac{j(X_{L2} - X_{C2})}{R + j(X_{L2} - X_{C2})} = 1.88 \angle 36.87^\circ V$

综上 $U_0 = 7.85 + 2.66 \sin(3\omega t + 36.87^\circ) V$

2.6.1

由换路定律: $i_1(0^-) = i_1(0^+)$ $i_2(0^-) = i_2(0^+)$ $U_C(0^+) = U_C(0^-)$

$\therefore i_1(0^+) = U_S / (R_1 + R_2) = 0.8mA$

$U_C(0^+) = U_C(0^-) = i_1(0^-) \cdot R_1 = 8V$

$i_2(0^+) = i_1(0^+) + i_2(0^+)$

$\therefore i_2(0^+) = -0.8mA$

$\therefore U_L(0^+) - U_C(0^+) + R_1 i_1(0^+) - R_2 i_2(0^+) = 0$

$\therefore U_L(0^+) = -4V$

2.6.4.

S合上前:

$U_C(0^-) = I_S \cdot R_S = 12V$

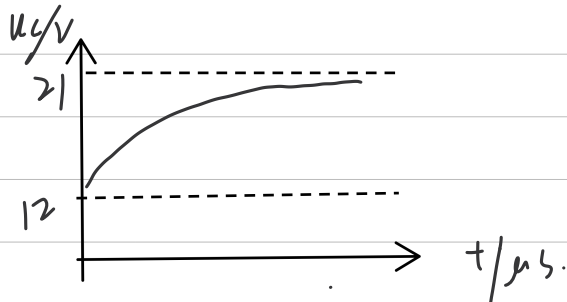
由换路定律: $U_C(0^+) = U_C(0^-) = 12V$

S合上后: U_C 的稳态值: $U_C = U_C(\infty) + [U_C(0^+) - U_C(\infty)] \cdot e^{-\frac{t}{\tau}}$

$\frac{U_C(\infty)}{R_S} = I_S + \frac{U_S - U_C(\infty)}{R_0} \quad \therefore U_C(\infty) = 21V$

$\tau = (R_0 + R_S) \cdot C = 3\mu s$

$\therefore U_C(t) = 12 - 9e^{-\frac{t}{3}} V$



2.6.6

当 S 闭合前:

$$i(0^-) = I_s = 11 \text{ A}.$$

由换路定律:

$$i(0^+) = i(0^-) = 11 \text{ A}.$$

当 S 闭合后:

$$\tau = \frac{L}{R} = 0.5 \text{ s} \quad i(\infty) = \frac{U_s}{R} + I_s = 16 \text{ A}.$$

$$\begin{aligned} i(t) &= i(\infty) + [i(0^+) - i(\infty)]e^{-\frac{t}{\tau}} = 16 + [11 - 16]e^{-2t} \\ &= 16 - 5e^{-2t} \text{ A} \end{aligned}$$